

CRYPTOGRAPHY ASSIGNMENT

① Cipher text: H L D T Q I H J O X W Q C M A G
 key = $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$ use A=0, B=1 ... find plaintext

→ find determinant:

$$|K| = (3)(7) - (2)(5) \\ = 21 - 10 = 11$$

$$11^{-1} \bmod 26 = 19$$

→ find inverse matrix:-

$$K^{-1} = 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \bmod 26 \Rightarrow \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

HL:

$$C = \begin{bmatrix} 7 \\ 11 \end{bmatrix}$$

$$P = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 11 \end{bmatrix} = \begin{bmatrix} 21 + 154 \\ 63 + 55 \end{bmatrix} = \begin{bmatrix} 175 \\ 118 \end{bmatrix}$$

$$175 \bmod 26 = 14 = T$$

$$118 \bmod 26 = 14 = O$$

DT: $C = \begin{bmatrix} 3 \\ 19 \end{bmatrix}$

$$P = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 19 \end{bmatrix} = \begin{bmatrix} 9 + 266 \\ 27 + 95 \end{bmatrix} = \begin{bmatrix} 275 \\ 122 \end{bmatrix}$$

$$275 \bmod 26 = 15 = P$$

$$122 \bmod 26 = 18 = S$$

$$RI \quad C = \begin{bmatrix} 16 \\ 8 \end{bmatrix}$$

$$P_2 \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 16 \\ 8 \end{bmatrix} = \begin{bmatrix} 160 \\ 184 \end{bmatrix}$$

$$160 \bmod 26 = 4 = E$$

$$184 \bmod 26 = 2 = C$$

$$HJ \quad C = \begin{bmatrix} 7 \\ 9 \end{bmatrix}$$

$$P_2 \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 9 \end{bmatrix} = \begin{bmatrix} 147 \\ 108 \end{bmatrix}$$

$$147 \bmod 26 = 17 = R$$

$$108 \bmod 26 = 4 = E$$

$$DX \quad C = \begin{bmatrix} 3 \\ 23 \end{bmatrix}$$

$$P_2 \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 23 \end{bmatrix} = \begin{bmatrix} 331 \\ 142 \end{bmatrix}$$

$$331 \bmod 26 = 19 = T$$

$$142 \bmod 26 = 12 = M$$

$$WQ : \quad C = \begin{bmatrix} 22 \\ 16 \end{bmatrix}$$

$$P_2 \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 22 \\ 16 \end{bmatrix} = \begin{bmatrix} 290 \\ 278 \end{bmatrix}$$

$$290 \bmod 26 = 14 = E$$

$$278 \bmod 26 = 18 = S$$

$$CM \quad C = \begin{bmatrix} 29 \\ 12 \end{bmatrix}$$

$$P_2 \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 29 \\ 12 \end{bmatrix} = \begin{bmatrix} 174 \\ 78 \end{bmatrix}$$

$$174 \bmod 26 = 18 = S$$

$$78 \bmod 26 = 0 = A$$

A_L

$$C = \begin{bmatrix} 0 \\ 6 \end{bmatrix}$$

$$P = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 6 \end{bmatrix} = \begin{bmatrix} 84 \\ 30 \end{bmatrix}$$

$$84 \bmod 26 = 6 = G$$

$$30 \bmod 26 = 4 = E$$

→ Top Secret message

② Consider Hill cipher key matrix $\begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix}$

cipher text: R X A O D Y find plain text for above cipher text

→ determinant:

$$|K| = 24 \times 14 - 19 \times 5$$

$$= 336 - 95 = 241$$

$$\rightarrow 241 \bmod 26 = 7$$

$$7^{-1} = 15$$

inverse matrix

$$K^{-1} = 15 \begin{bmatrix} 14 & -19 \\ -5 & 24 \end{bmatrix} \bmod 26$$

$$K^{-1} = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix}$$

R X

$$\begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 17 \\ 23 \end{bmatrix} = \begin{bmatrix} 57 \\ 557 \end{bmatrix}$$

$$57 \bmod 26 = 5 = F$$

$$557 \bmod 26 = 11 = L$$

$$A_0 \quad C_2 \begin{bmatrix} 0 \\ 14 \end{bmatrix}$$

$$P_2 \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 0 \\ 14 \end{bmatrix} = \begin{bmatrix} 14 \\ 308 \end{bmatrix}$$

$$14 \bmod 26 = 14 \rightarrow D$$

$$308 \bmod 26 = 22 \rightarrow W$$

Q4

$$C_2 \begin{bmatrix} 3 \\ 24 \end{bmatrix}$$

$$P_2 \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 3 \\ 24 \end{bmatrix} = \begin{bmatrix} 30 \\ 537 \end{bmatrix}$$

$$30 \bmod 26 = 4 \rightarrow E$$

$$537 \bmod 26 = 17 \rightarrow R$$

FLOWER

④ Transposition encryption

encrypt "The arrow is pointing north direction"
using simple with key = 4

THEA

RROW

ISPO

INTI

NLND

RTHD

IREL

TIDN

Cipher text : TRIINRITHRSNLTRIEOPINHEOAW
DIDREN

→ keyword

using key = ATLAS

① ⑤ ③ ② ④
A T L A S
T H E A R
R O W I S
P O I N T
I N G N O
R T H D I
R E C T I
O N X X X

= T R P I R R O A I N N D T X L E W I L H C X R S T O I I X H O
ONTEN

double & key : key 1 = Break
key 2 = log

② ⑤ ③ ② ④
B R E A K
T H E A R
R O W I S
P O I N T
I N G N O
R T H D I
R E C T I
O N X X X

A I N N D T X T R P I R R O E W I L H C X R S T O I I X
H O O N T E N

key 2: LOH

L O G

② ③ ①

A T N

N D T

T R P

I R R

O E W

I G WH

C R S

T O I

H O O

N T E

N + +

NTPRWHSTIOEANTI OICTHNNIDRRLEROOT

→ decryption with key 10 (simple):

TIDIE MRANME RED40 SANDW RSLTP EED0
SSANO AHT00 BIENP LODAE PEIXD ELNSX

$$\begin{aligned}\text{key} &= 10, \text{ character} = 60 \\ &= 60 / 10 = 6\end{aligned}$$

T R O O O P S H E A P
I N G W E S T N E E
D M O R E S U P P L
I E S S E N D G E N
E R A L D U B O I S
M E N T O A I O X X

= TROOPS HEADING WEST NEED MORE SUPPLIES
SEND GENERAL DUBOIS TO AID

2) Cipher: ILEDT WHPTR SOIIT MAXSA XERXT

ANONI SNFOTX

Keyword: SECRET

521436

~~36/2~~

$$36/2 = 6$$

S E C R E T

S 2 1 4 3 6

T H I S I S

A P L A I N

N T E X T F

O R D E M O

N S T R A T

I O N X X X

This is a plain text for demonstration^{xxx}

⇒ Keyword - PLAIN

Cipher - VEX EAX ARA ATX HGD

PLAIN ⇒ ALLNP

$$15/5 = 3$$

P L A I N

H A V E A

G R E A T

D A Y X X

⇒ Have a great day

⇒ permutation - find inverse permutation of

(1) 4 3 1 5 2
① ② ③ ④ ⑤

$$⇒ P^{-1} = \underline{\underline{3, 5, 2, 1, 4}}$$

②

7	4	1	3	5	2	6
1	2	3	4	5	6	7

$P^{-1} = 3, 6, 4, 2, 5, 7, 1$

③

4	1	3	2
①	②	③	④

$P^{-1} = 2, 4, 3, 1$

④

3	5	1	4	2
1	2	3	4	5

$P^{-1} = \underline{3, 5, 1, 4, 2}$

confusion vs. Diffusion:

Analyse the following classical ciphers with respect to confusion and diffusion.

Complete the table and justify your classification based on how each cipher transforms the plaintext.

Cipher	Does it change the character/symbol value?	Does it change the character position?	Confusion / Diffusion	Justification
Shift Cipher	Yes	No	Confusion	Each plaintext character is replaced by another character by shifting its value. The position of characters remains unchanged.
Multiplicative Cipher	Yes	No	Confusion	Each character value is multiplied by a key modulo 26. The character positions are not changed.
Affine Cipher	Yes	No	Confusion	Each character value is transformed using $C = (aP + b) \bmod 26$. The positions of characters remain unchanged.
Transposition Cipher	No	Yes	Diffusion	The character values remain the same, but their positions are rearranged according to a key or transposition rule.